

HW04 – Automation Testing

Playwright AI-First Automation Report

1. Introduction

1.1 Purpose

This assignment applies an AI-first approach to UI test automation. AI tools were used to analyze the system under test, understand the existing test cases and API requirements, identify frontend selectors and backend endpoints, generate test data, and produce Playwright automation scripts.

The generated scripts were subsequently reviewed and corrected manually. The final automation suite is therefore the responsibility of the student and includes positive, negative, boundary-value, data-driven, and multi-browser testing.

1.2 Student and Project Information

Item	Information
Student ID	23127325
Course	Software Testing
Assignment	HW04 – Automation Testing
Framework	Playwright
Test Data	JSON
Features	FR-03, FR-11, FR-13
Browsers	Chromium, Firefox, Edge

1.3 Features Automated

- FR-03 — Forgot Password & Reset
- FR-11 — Order History
- FR-13 — Dashboard Metrics

2. AI-First Generation Process

1. AI reads testcases.md
2. AI reads API requirements
3. AI inspects frontend components
4. AI identifies routes and selectors
5. AI inspects backend APIs requ
6. AI inspects seed data
7. AI maps test cases to implementation
8. AI generates JSON test data
9. AI generates Playwright scripts
10. Human reviews and fixes scripts
11. Execute Chromium / Firefox / Edge
12. Analyze failures and report defects

2.1 Reading Test Case Design

- Test case IDs

- Positive and negative scenarios
- Boundary-value cases
- Expected results
- Preconditions
- Required test data
- Feature-specific behavior

The AI used the test-case design as the basis for mapping each requirement to an executable Playwright scenario.

2.2 Reading API Requirements

The AI inspected API requirements because FR-03 authentication/reset behavior, FR-11 order history, and FR-13 dashboard metrics depend on backend state.

2.3 Inspecting Frontend Components

- Login forms
- Navigation elements
- Tables
- Buttons
- Headings
- Input fields
- Status labels
- Dashboard statistic cards
- Conditional rendering
- Routes

2.4 Inspecting Seed Data

Seed data was inspected to identify valid users, roles, orders, and other entities required by the tests. Controlled data was preferred where specific backend state was required.

3. Test Automation Coverage

3.1 Summary

Feature	Spec	Cases	Browsers	Total Runs
FR-03 — Forgot Password & Reset	scripts/fr03-forgot-password.spec.js	19	Chromium, Firefox, Edge	57
FR-11 — Order History	scripts/fr11-order-history.spec.js	19	Chromium, Firefox, Edge	57
FR-13 — Dashboard Metrics	scripts/fr13-dashboard.spec.js	17	Chromium, Firefox, Edge	51
Total		55	3	165

55 test cases × 3 browsers = 165 browser runs.

3.2 FR-03 — Forgot Password & Reset

Automation: scripts/fr03-forgot-password.spec.js

Data: data/fr03-forgot-password.json

19 cases covering registered/unregistered email, empty and malformed input, password boundaries, OTP validation, password requirements, confirmation, successful reset, and OTP expiration.

ID	Type	Description
TC-FR03-001	Positive	Submit a registered email → OTP form (step 2)
TC-FR03-002	Negative	Syntactically valid but unregistered email → error alert
TC-FR03-003	Negative	Empty email field → submit blocked (HTML5 required)
TC-FR03-004	Negative	Malformed email (no @) → backend rejects as user-not-found
TC-FR03-005	Negative	Whitespace-only email → backend rejects as user-not-found
TC-FR03-006	Edge (BVA)	Exactly 8-char valid password (success), uses fresh user
TC-FR03-007	Negative	Incorrect OTP (4 digits) with a strong password → error alert
TC-FR03-008	Negative	Empty OTP field → submit blocked (HTML5 required)
TC-FR03-009	Edge (BVA)	Password shorter than 8 chars, just below minimum length → error alert
TC-FR03-010	Negative	Password missing whitespace → SUT regex rejects
TC-FR03-011	Negative	Password missing uppercase letter → error alert
TC-FR03-012	Negative	Password missing lowercase letter → error alert
TC-FR03-013	Negative	Password missing digit → error alert
TC-FR03-014	Positive	Successful full reset flow with strong password → login redirect

TC-FR03-015	Negative	Empty new-password field → submit blocked (HTML5 <i>required</i>)
TC-FR03-B001	Edge (BVA)	OTP length must equal 6 (spec requirement)
TC-FR03-B002	Edge (BVA)	Step-2 must expose a confirm-password input next to its label
TC-FR03-B003	Edge (BVA)	Password <code>Password1!</code> (10 chars, upper+lower+digit+special) accepted
TC-FR03-B004	Negative	Reset token rejected after a long wait (~5 min TTL)

3.3 FR-11 — Order History

Automation: `scripts/fr11-order-history.spec.js`

Data: `data/fr11-order-history.json`

19 cases covering empty/non-empty history, sorting, currency formatting, order statuses, cancel behavior, authentication, boundary amounts, and user isolation.

ID	Type	Description
TC-FR11-001	Edge	Authenticated user with no orders sees the empty-state message
TC-FR11-002	Edge	Table header <code>Lịch sử đơn hàng</code> rendered when logged in
TC-FR11-003	Positive	Table renders exactly 1 row when the user has 1 order
TC-FR11-004	Edge (BVA)	Table renders exactly 5 rows (BVA: low-N boundary)
TC-FR11-005	Edge	Orders sorted newest-first (descending id) when seeded ascending
TC-FR11-006	Edge	Amount rendered with <code>₫</code> suffix and locale-grouped digits (<code>100.000 ₫</code>)

TC-FR11-007	Edge	pending badge shows Chờ xác nhận with yellow bg
TC-FR11-008	Edge	confirmed badge shows Đã xác nhận with indigo bg
TC-FR11-009	Edge	shipping badge shows Đang giao with blue bg
TC-FR11-010	Edge	delivered badge shows Đã giao with green bg
TC-FR11-011	Edge	canceled badge shows Đã hủy with red bg
TC-FR11-012	Edge	Cancel button visible on a pending row
TC-FR11-013	Edge	Cancel button NOT rendered on a delivered row
TC-FR11-014	Edge	Cancel button NOT rendered on a canceled row
TC-FR11-015	Positive	Click Hủy đơn on a pending order → success alert + status flips to canceled
TC-FR11-016	Negative	Unauthenticated visit to /profile renders the "please log in" message
TC-FR11-017	Edge (BVA)	A 1-VND amount formats as 1 đ with the đ suffix
TC-FR11-018	Edge (BVA)	A 9-digit amount formats with two grouping separators
TC-FR11-019	Negative	Isolation: another user's order does NOT appear in this user's table

3.4 FR-13 — Dashboard Metrics

Automation: scripts/fr13-dashboard.spec.js

Data: data/fr13-dashboard.json

17 cases covering authentication, dashboard rendering, order counts, revenue calculations, boundary amounts, role checks, navigation, formatting, and detection of the revenue calculation defect.

ID	Type	Description
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TC-FR13-001	Negative	Unauthenticated visit renders the Admin Login form instead of the dashboard
TC-FR13-002	Positive	Successful login with seeded admin credentials renders the dashboard tab
TC-FR13-003	Edge	Dashboard header Dashboard is rendered after login
TC-FR13-004	Edge	Tổng doanh thu (Delivered) stat card is rendered
TC-FR13-005	Edge	Tổng số đơn hàng stat card is rendered
TC-FR13-006	Positive	Adding N pending orders increases order count by exactly N
TC-FR13-007	Positive	Adding N pending orders does NOT change delivered revenue
TC-FR13-008	Positive	Adding 1 delivered order increases displayed revenue by total amount
TC-FR13-009	Edge (BVA)	BVA: 1-VND delivered order → revenue increases by 1 (SUT defect)
TC-FR13-010	Edge (BVA)	BVA: 30,000,000-VND delivered order → revenue increases by 30,000,000
TC-FR13-011	Negative	Non-admin login shows the Bạn không phải là admin! alert
TC-FR13-012	Negative	Wrong admin password shows the Đăng nhập thất bại alert
TC-FR13-013	Negative	Before login, sidebar nav items are NOT rendered
TC-FR13-014	Edge	After login, sidebar nav items are rendered (Dashboard tab visible)
TC-FR13-015	Positive	Click Đơn hàng sidebar tab → switches away from dashboard

TC-FR13-016	Edge	Revenue cell formatted with Vietnamese grouping and a đ suffix
TC-FR13-017	Bug	BUG-FR-13-001 detector — dashboard revenue must equal the correct sum of non-pending amounts

4. Data-Driven Automation Design

Test data is stored separately from the Playwright implementation.

data/

└─ fr03-forgot-password.json

└─ fr11-order-history.json

└─ fr13-dashboard.json

scripts/

└─ fr03-forgot-password.spec.js

└─ fr11-order-history.spec.js

└─ fr13-dashboard.spec.js

- Adding a test case is primarily a JSON-data change.
- Test data is not hardcoded as inline arrays or objects in the specification.
- The same test logic is reused across browsers.
- Test cases can be traced back to testcases.md.

4.1 Data Seeding

Controlled seed data is used where a test requires a known backend state, especially for order history, dashboard order counts, dashboard revenue, isolation, and boundary-value amounts.

5. Multi-Browser Execution

Feature	Chromium	Firefox	Edge
FR-03	✓	✓	✓
FR-11	✓	✓	✓
FR-13	✓	✓	✓

FR-03: 57 runs; FR-11: 57 runs; FR-13: 51 runs; total: 165 browser runs.

Note: Playwright WebKit is not Edge. Edge uses the Chromium engine. This report documents the browsers actually used: Chromium, Firefox, and Edge.

6. Automation Results / Playwright HTML Reports

Playwright HTML reports were generated for browser execution. Reports provide test case, browser, status, duration, assertion failure, locator, and call-log information, with screenshots or traces where configured.

Each report visibly identifies the student: Run by: 23127325

Feature	Cases	Browser Runs	Report
FR-03	19	57	Playwright HTML
FR-11	19	57	Playwright HTML
FR-13	17	51	Playwright HTML
Total	55	165	HTML reports

Failures were manually investigated. FR-11 failures exposed a locale/separator expectation problem, while an FR-13 failure exposed an ambiguous selector rather than a missing Dashboard element.

6.1 Overall Results

Result	Count	Percentage
Total tests	165	100%
Passed	150	91%
Failed	15	9%
Flaky	0	0%
Skipped	0	0%

Q

Search tests

All

165

✓ Passed

150

✗ Failed

15

Flaky

0

Skipped

0

> Metadata

8/9/2026, 9:23:33 PM Total time: 3.0m

HW04 EShop Automation — Run by: 23127325

▼ fr03-forgot-password.spec.js

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B001 — OTP length must equal 6 (spec requirement)

1.9s

chromium

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B002 — step-2 must expose a confirm-password input next to its label

8.8s

chromium

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B003 — password 'Password1!' (10 chars, upper+lower+digit+special) must be accepted

3.7s

chromium

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B004 — resetToken must be rejected after a long wait (~5 min) — reset must NOT succeed

6.6s

chromium

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B001 — OTP length must equal 6 (spec requirement)

1.8s

edge

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B002 — step-2 must expose a confirm-password input next to its label

9.2s

edge

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B003 — password 'Password1!' (10 chars, upper+lower+digit+special) must be accepted

5.3s

edge

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

✗ FR-03 Forgot Password & Reset — Run by: 23127325 › TC-FR03-B004 — resetToken must be rejected after a long wait (~5 min) — reset must NOT succeed

6.7s

edge

fr03-forgot-password.spec.js:196 ▶️ 📄 View Trace

6.2 Bugs

1. TC-FR03-B001 — OTP length validation

- Issue: OTP length does not satisfy the required 6-digit specification.
- Expected: The system should require an OTP with exactly 6 digits.
- Actual: The test detected that the OTP validation does not correctly enforce the 6-digit requirement.
- GitHub Issue: #4
- Status: Open

2. TC-FR03-B002 — Missing confirm-password field

- Issue: The reset-password Step 2 does not expose the required confirm-password input.
- Expected: Step 2 should contain a password confirmation field next to its label.
- Actual: The required confirm-password input is missing.
- GitHub Issue: #2
- Status: Open

3. TC-FR03-B003 — Password1! rejected

- Issue: A password that satisfies the specified password requirements is rejected.
- Expected: Password1! should be accepted because it contains uppercase, lowercase, digit, and special character and has sufficient length.
- Actual: The password is rejected.
- GitHub Issue: #3
- Status: Open

4. TC-FR03-B004 — Reset token expiration

- Issue: The reset token is not correctly rejected after the specified expiration period.
- Expected: A resetToken that has expired after approximately 5 minutes should be rejected and the password reset should not succeed.
- Actual: The token expiration behavior does not satisfy the requirement.
- GitHub Issue: #1
- Status: Open

7. AI-Generated Script Review / Gap Analysis

7.1 FR-11 — Incorrect Handling of Thousands Separators

1. Hardcoded thousands separator

- AI issue: Expected currency values were hardcoded with '.' as the thousands separator, for example 100.000 ¢ and 1.234.567.890 ¢.
- Evidence: TC-FR11-006 expected 100.000 ¢ but received 100,000 ¢. TC-FR11-018 expected 1.234.567.890 ¢ but received 1,234,567,890 ¢.
- Why AI missed it: The AI assumed the separator specified in the test case was fixed and did not sufficiently consider locale/browser-dependent formatting.
- Human review/fix: The application formatting was reviewed and the assertion was revised to avoid an unjustified dependency on one separator or to explicitly enforce the required locale.
- Impact: The automation produced false failures even though the numeric values and ¢ suffix were correct.

7.2 FR-13 — Ambiguous Dashboard Text Selector

1. Ambiguous Dashboard text selector

- AI issue: AI used page.getByText('Dashboard') to verify the Dashboard page.
- Evidence: TC-FR13-002 failed because the locator matched both the sidebar Dashboard item and the Dashboard <h2> heading.
- Why AI missed it: AI assumed the text Dashboard uniquely identified the target element and did not sufficiently inspect duplicate text in the DOM.
- Human review/fix: Replaced the generic text locator with a semantic heading locator.
- Impact: The test failed with a Playwright strict-mode violation even though the Dashboard was correctly rendered.

Corrected assertion:

```
await expect(page.getByRole('heading', { name: 'Dashboard', exact: true })).toBeVisible();
```

8. Human Corrections and Final Validation

8.1 Selector Review

- Checked selectors against the actual frontend.
- Removed ambiguous generic selectors where necessary.
- Preferred semantic Playwright locators.

8.2 Assertion Review

Assertions were reviewed to verify actual expected behavior rather than merely confirming that a page loaded.

8.3 Test Data Review

- Required fields were present.
- Positive, negative, and boundary cases were represented.
- Data matched actual application behavior.
- Cases were reproducible.

8.4 Cross-Browser Validation

- Chromium
- Firefox
- Edge

Failures were investigated to distinguish SUT defects from incorrect expectations, browser behavior, locator problems, and test-data problems.

8.5 Final Responsibility

Although AI generated the initial test data and automation scripts, the final scripts were reviewed, corrected, and validated manually. The AI-generated implementation was treated as a starting point rather than the final source of truth.

8. AI Critique

The AI performed well in generating the overall structure of the automation tests, including data-driven test cases, Playwright scripts, selectors, API-based test-data setup, and reusable testing patterns. However, it was not completely reliable when decisions required detailed inspection of the actual application behavior. One concrete issue was the use of `page.getByText('Dashboard')` with `toBeVisible()`. The Dashboard text appeared in both the sidebar and the main `<h2>` heading, causing Playwright strict-mode failure because the locator resolved to multiple elements. The AI failed to anticipate this ambiguity because it treated visible text as a unique identifier rather than checking how the same text was represented across the DOM. The AI also initially relied on assumptions about currency formatting instead of fully accounting for locale-dependent thousands separators. These issues demonstrate that AI can generate syntactically valid and logically plausible tests while still missing implementation-specific details. The main principle I learned is that AI should be treated as a test-generation assistant, not an autonomous tester. Human review is necessary to verify selectors against the actual DOM, confirm API behavior, inspect seed data, and execute the generated tests. The most reliable workflow is AI generation → source-code verification → test execution → human review → correction → final execution.

9. Conclusion

The HW04 automation suite demonstrates an AI-first approach to Playwright test automation across FR-03, FR-11, and FR-13. A total of 55 test cases were automated and executed across Chromium, Firefox, and Edge, resulting in 165 browser runs.

The automation uses separate JSON test-data files and data-driven Playwright specifications and covers positive, negative, and boundary-value scenarios.

Human review was essential. The review identified ambiguous selectors, overly strict assumptions about number formatting, and insufficient validation of dashboard calculations and boundary conditions.

The final scripts were reviewed and corrected before being used as the submitted automation suite.

10. Self-grade

No.	Criteria	Grade	Self-Assessed Grade
1	Task 1 - Feature A	25	25
1	Task 1 - Feature B	25	25
1	Task 1 - Feature C	25	25
2	Task 2 — Demo video	15	15
3	Agent Skills	10	10
	Total	100	100